

Chapter 9:

Infinite Series

Section 9.5: Alternating Series

An **alternating series** is a series whose terms are alternately positive and negative. Here are two examples:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$$

$$-\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \frac{6}{7} - \cdots = \sum_{n=1}^{\infty} (-1)^{n} \frac{n}{n+1}$$

We see from these examples that the nth term of an alternating series is of the form

$$\sum_{n=1}^{\infty} (-1)^n a_n \qquad or \qquad \sum_{n=1}^{\infty} (-1)^{n-1} a_n \quad or \quad \sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

where a_n is positive for all n.

THEOREM 9.14 Alternating Series Test

Let $a_n > 0$. The alternating series

$$\sum_{n=1}^{\infty} (-1)^n a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converge when these two conditions are met.

1. $\lim_{n\to\infty} a_n = 0$
2. $a_{n+1} \leq a_n$, for all n



Remark:

If the limit $\lim_{n\to\infty} a_n = c \neq 0$ or $\lim_{n\to\infty} a_n = \infty$ or $-\infty$ for an alternating series, then the series diverges by the *n*th-Term Test for Divergence.

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

This series is called **alternating harmonic series**

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{\pi}{n}\right)$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{n\cos(n\pi)}{2^n}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^n \frac{3\sqrt{n+1}}{\sqrt{n}+1}$$

Estimating Sums

For a convergent alternating series, the partial sum S_n can be a useful approximation for the sum S of the series. The error involved in using $S \approx S_n$ is the remainder $R_n = S - S_n$.

THEOREM 9.15 Alternating Series Remainder

If a convergent alternating series satisfies the condition $a_{n+1} \leq a_n$, then the absolute value of the remainder R_N involved in approximating the sum S by S_N is less than (or equal to) the first neglected term. That is,

$$|S - S_N| = |R_N| \leq a_{N+1}.$$

A proof of this theorem is given in Appendix A.



EXAMPLE

Finding the Number of Terms In Exercises 35–40, use Theorem 9.15 to determine the number of terms required to approximate the sum of the series with an error of less than 0.001.

37. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n^3 - 1}$

$n = 7$

EXAMPLE

The given alternating series is convergent by the AST. How many terms of the series do we need to add in order to find the sum to the indicated accuracy?

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{5\sqrt{n+3}}, \qquad |error| < 0.05$$

$n = 13$

Absolute and Conditional Convergence

THEOREM 9.16 Absolute Convergence

If the series $\Sigma\,|a_n|$ converges, then the series $\Sigma\,a_n$ also converges.



EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$$

Notice that this series is NOT alternating series.

Absolute and Conditional Convergence

THEOREM 9.16 Absolute Convergence

If the series $\Sigma |a_n|$ converges, then the series Σa_n also converges.



The converse of Theorem 9.16 is not true. For instance, the **alternating harmonic series**

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

converges by the Alternating Series Test. Yet the harmonic series diverges. This type of convergence is called **conditional**.

Definitions of Absolute and Conditional Convergence

1. The series Σa_n is **absolutely convergent** when $\Sigma |a_n|$ converges.
2. The series Σa_n is **conditionally convergent** when Σa_n converges but $\Sigma |a_n|$ diverges.

EXAMPLE

Determine whether the series converges absolutely or conditionally, or diverges.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

EXAMPLE

Determine whether the series converges absolutely or conditionally, or diverges.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n\sqrt{n}}$$

EXAMPLE

Determine whether the series converges absolutely or conditionally, or diverges.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\, n}{5n+1}$$

Rearrangement of Series

A finite sum such as

$$1 + 3 - 2 + 5 - 4$$

can be rearranged without changing the value of the sum. This is not necessarily true of an infinite series—it depends on whether the series is absolutely convergent or conditionally convergent.

1. If a series is *absolutely convergent*, then its terms can be rearranged in any order without changing the sum of the series.

2. If a series is *conditionally convergent*, then its terms can be rearranged to give a different sum.

The second case is illustrated in Example 8, page #628. Please read it.

Solve exercises of section 9.5.

Start with the suggested problems mentioned in the syllabus.